Insurance Charges Prediction System

1. Problem Statement

Domain: Machine Learning

Learning: Supervised Learning

Algorithm: Multiple Linear Regression

Reason: We have multiple independent variables (age, sex, BMI, children, and smoker) and one dependent variable (charges). Therefore, Multiple Linear Regression is suitable for this prediction problem.

2. Basic Information About the Dataset

- **Number of Rows:** 1,338
- **Number of Columns:** 6

3. Pre-processing Methods

Convert categorical values into numerical values using Label Encoding.

- **Sex:** female → 0, male → 1
- **Smoker:** no → 0, yes → 1

4. Model Creation Phase

The best-performing model was created using the **Random Forest Regression algorithm**, which achieved an **R^2 score of 0.87**.